

Title

Prophylactic Vacuum-Assisted Wound Closure in Emergency Abdominal Surgery: Study Protocol for a Multicenter Randomized Trial

Abstract

Surgical site infection remains a substantial source of postoperative morbidity and healthcare expenditure despite modern aseptic techniques. While negative pressure wound therapy has demonstrated efficacy in managing established wound complications, evidence regarding its prophylactic application in high-risk emergency surgical populations remains limited. This multicenter protocol describes a randomized trial comparing prophylactic negative pressure wound management with standard dressing in patients undergoing emergency abdominal surgery. Ninety participants will be enrolled and randomly assigned to either five days of prophylactic vacuum therapy ($n = 45$) or conventional postoperative wound dressing management ($n = 45$). Primary outcome measures superficial and deep surgical site infections diagnosed through standardized criteria within 30 postoperative days. Secondary measures include cost-effectiveness, wound healing progression, and patient satisfaction. This study addresses whether preventive vacuum application reduces infection risk in this vulnerable surgical population.